P10 (ii) Deduce that

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i) - \frac{2}{n} \sum_{1 \leq i_1 < i_2 \leq n} P(A_{i_1} \cap A_{i_2})$$

Note that $P\left(\bigcup_{i=1}^{n} A_{\sigma(i)}\right) \leq \sum_{i=1}^{n} P(A_{\sigma(i)}) - \sum_{i=1}^{n-1} P(A_{\sigma(i)} \cap A_{\sigma(i+1)})$

for any permutation σ on $\{1, 2, \dots, n\}$.

We see that $P\left(\bigcup_{i=1}^{n} A_{\sigma(i)}\right) = P\left(\bigcup_{i=1}^{n} A_i\right)$ and $\sum_{i=1}^{n} P(A_{\sigma(i)}) = \sum_{i=1}^{n} P(A_i)$.

~~Sum~~ Each permutation gives an ~~equal~~ inequality.

Sum them up we get

$$\underline{n!}\, P\left(\bigcup_{i=1}^{n} A_i\right) \leq \underline{n!} \sum_{i=1}^{n} P(A_i) - \underline{\sum_{\sigma}} \sum_{i=1}^{n-1} P(A_{\sigma(i)} \cap A_{\sigma(i+1)})$$

There are $n!$ permutations.

Sum over all permutations σ.

Fix $1 \leq i_1 < i_2 \leq n$.

We want to count the number of $P(A_{i_1} \cap A_{i_2})$ appearing in $\sum_\sigma \sum_{i=1}^{n-1} P(A_{\sigma(i)} \cap A_{\sigma(i+1)})$

Since $P(A_{i_1} \cap A_{i_2})$ appears at most once in $\sum_{i=1}^{n-1} P(A_{\sigma(i)} \cap A_{\sigma(i+1)})$ for any permutation σ, we can count the permutations σ such that $\{\sigma(i), \sigma(i+1)\} = \{i_1, i_2\}$ for some $i = 1, \dots, n-1$.

$\#$ perm. with this property $= (n-1) \cdot 2 \cdot (n-2)! = 2(n-1)!$

Here $n-1$ is $\#$ ways to choose an i for $\{\sigma(i), \sigma(i+1)\}$,

2 is $\#$ ways to assign i_1, i_2 to $i, i+1$,

$(n-2)!$ is $\#$ ways to assign remaining numbers $\{1, \dots, n\} \setminus \{i_1, i_2\}$ to the numbers in $\{1, 2, \dots, n\} \setminus \{i, i+1\}$.

$\therefore$ ~~Each~~ $P(A_{i_1} \cap A_{i_2})$ appears $2(n-1)!$ times in $\sum_\sigma \sum_{i=1}^{n-1} P(\cdots)$.

$$\therefore \sum_{\sigma} \sum_{i=1}^{n-1} P\left(A_{\sigma(i)} \cap A_{\sigma(i+1)}\right) = \sum_{1 \le i_1 < i_2 \le n} 2(n-1)!\, P(A_{i_1} \cap A_{i_2})$$

$$\therefore \quad n!\, P\left(\bigcup_{i=1}^{n} A_i\right) \le n! \sum_{i=1}^{n} P(A_i) - 2(n-1)! \sum_{1 \le i_1 < i_2 \le n} P(A_{i_1} \cap A_{i_2})$$

$$\Rightarrow \quad P\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} P(A_i) - \frac{2}{n} \sum_{1 \le i_1 < i_2 \le n} P(A_{i_1} \cap A_{i_2}).$$